

Immersive VR/AR safety training with biometric feedback for evaluating worker readiness in high-risk construction zones.

Construction × Immersive workforce training × AR / VR / XR · OS114 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
23 is the world moving	13 can we play, can we win	LOW 0 named in the evidence	€5.9m partial evidence · per year	LATER research and standards activi...	L3 10 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 1.0 published reference(s) in Construction, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is an immersive VR/AR safety training solution with biometric feedback for construction workers in high-risk zones, enabling evaluation of worker readiness and behavioral change. It targets construction firms and safety officers who need to move beyond traditional training to measurable, behavior-level outcomes. The need is urgent because research shows unsafe behaviors remain a leading cause of accidents, and current evaluation methods struggle to measure knowledge transfer and behavioral change.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€11.6m €609k – €56.5m	€5.9m €306k – €28.4m	€7k €383 – €36k	partial
Observed: tendered contract value, annualised	€123m €123m – €471m	€0	€0	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Construction (PT+SE)	15,256 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting AR / VR / XR	0.8 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_ain2 · E_AI_TIR	2025
Annual value of one engagement	€952k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,540 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Artificial intelligence by NACE Rev. 2 activity series E_AI_TIR is used as a proxy for AR / VR / XR: No XR series published; image processing as nearest. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 4 declared geographies are outside the European reference data (BR, US); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

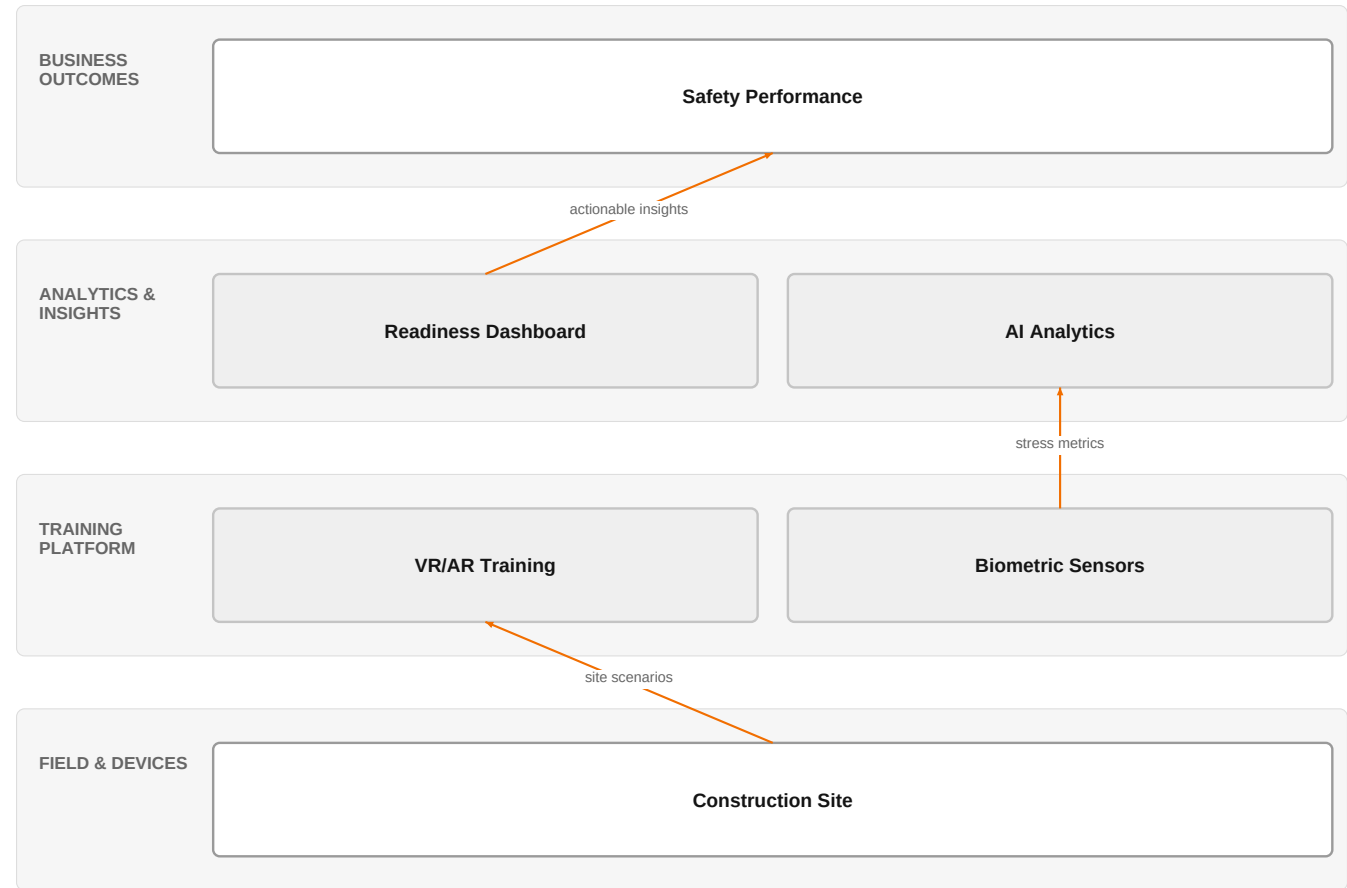
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€123m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07

Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a
--	--------------	------------	---------------------------------------	----------------

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 24 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 4 declared geographies are outside the European reference data (BR, US); the estimate covers the European footprint only.

The solution, in one picture

Immersive Safety Training with Biometric Feedback



Orange asset Partner Customer-owned Third party / to be sourced

This solution integrates VR/AR training with biometric feedback to provide measurable readiness and safety outcomes.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent research highlights that immersive technologies (VR/AR) are being explored for evaluating safety training in high-risk environments like construction, addressing the challenge of measuring knowledge transfer and behavioral change (SIG-A33E4A343E60). Studies demonstrate that VR simulations combined with wearable sensors can assess workers' neuro-physiological stress responses to AR warnings, indicating a move toward biometric feedback in training (SIG-8BC657BF5DA5). Preliminary design guidelines are emerging for behavior-level evaluation in high-risk work, such as working at height (SIG-7ADD7236CDF2). Additionally, machine learning approaches are being applied to predict safety behaviors and health risks, suggesting a data-driven shift in safety management (SIG-9528B2CFF302, SIG-6ADB457BBC36).

Sources: SIG-A33E4A343E60 openalex.org 2025-12-17; SIG-8BC657BF5DA5 openalex.org 2025-10-04; SIG-7ADD7236CDF2 openalex.org 2025-09-01; SIG-9528B2CFF302 openalex.org 2026-01-01; SIG-6ADB457BBC36 openalex.org 2025-11-19

Who buys, and why now

The buyer is typically a Health & Safety Director or a Training Manager at a large construction firm, while the pain is felt by site supervisors who see recurring unsafe behaviors despite conventional training. The trigger is often a serious incident, a regulatory audit finding, or a client requirement for demonstrable safety competence. They need to prove that training translates into safe behavior on site, which current methods fail to do.

Why it is hot now — every claim bound to a dated source

- Immersive technologies offer opportunities for evaluating safety training in high-risk environments, addressing challenges in measuring knowledge transfer and behavioral change. [SIG-A33E4A343E60]
- Immersive VR simulations with wearable sensors can assess workers' neuro-physiological stress responses to augmented reality safety warnings in virtual roadway work zones. [SIG-8BC657BF5DA5]
- Preliminary design guidelines support the evaluation of industrial safety training using immersive technologies, focusing on behavior-level evaluation in high-risk work environments. [SIG-7ADD7236CDF2]
- Immersive technologies are being evaluated for safety training in high-risk environments like construction and mining, with a focus on measuring knowledge transfer and behavioral change. [SIG-A33E4A343E60]
- Design guidelines support evaluating immersive safety training, especially for behavior-level outcomes in high-risk work like working at height. [SIG-7ADD7236CDF2]
- Immersive VR simulations with AR warnings are used to assess construction workers' stress responses, indicating potential for safety training. [SIG-8BC657BF5DA5]
- Immersive technologies are being reviewed for evaluating industrial safety training in high-risk environments like construction. [SIG-7ADD7236CDF2, SIG-A33E4A343E60]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its AI / Data / Cloud experts and Digital experts to integrate VR/AR training platforms with biometric sensors, and leverage its AR / VR / XR technology capability. The engagement would start with a needs assessment and pilot design, then deploy the immersive training environment, integrate biometric data collection, and finally provide analytics dashboards for safety managers to track readiness and behavioral change.

Why Orange can win

Orange can win by combining its expertise in AI/Data/Cloud and digital transformation with its AR/VR/XR technology, offering an integrated solution that competitors may lack. However, the assets are generic; Orange does not have a dedicated construction safety product, so differentiation must come from its ability to orchestrate partners and deliver end-to-end managed services.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
AR / VR / XR	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

LOW competitive intensity (score 2.5; 0 of 7 listed players appear in this topic's own evidence)

One or two adjacent players. Orange would be shaping the space rather than fighting for it.

Competitor	Category	Position here	Basis	Evidence
PTC (ThingWorx)	Industrial / OT specialist	sells AR / VR / XR	structural	—
Econocom	Regional integrator or reseller	sells AR / VR / XR; competes in EX: Employee Experience	structural	—
Schneider Electric	Industrial / OT specialist	active in Construction	structural	—
Siemens	Industrial / OT specialist	active in Construction	structural	—
Microsoft	Hyperscaler	competes in EX: Employee Experience — also an Orange partner	structural	—
Bechtle	Regional integrator or reseller	competes in EX: Employee Experience	structural	—

Computacenter	Regional integrator or reseller	competes in EX: Employee Experience	structural	—
---------------	---------------------------------	-------------------------------------	------------	---

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently measure the effectiveness of your safety training beyond completion rates?
- 2 Have you experienced any recent incidents that you attribute to a gap between training and on-site behavior?
- 3 What is your current process for assessing worker readiness before they enter high-risk zones?
- 4 Are you already using any digital or VR-based training tools? If so, what are their limitations?
- 5 How would your safety team react to using biometric data (e.g., heart rate, stress levels) as part of training evaluation?
- 6 What is your budget and timeline for upgrading your safety training program?

Objections you will meet

They will say	You can answer
VR training is a gimmick; it won't change real behavior.	You're right that VR alone won't change behavior, but when combined with biometric feedback and data analytics, it can provide objective evidence of stress responses and readiness, which traditional training cannot. Our approach focuses on behavior-level evaluation, not just simulation.
We already have safety training; why invest in something unproven?	Your current training likely measures knowledge, not behavior. Research shows that measuring behavioral change is a key challenge. This solution addresses that gap, and we can start with a pilot to prove value before full deployment.
Workers won't wear biometric sensors; it's invasive.	Privacy is a valid concern. We design the solution with worker consent and data anonymization. The sensors are non-intrusive and used only during training sessions, not on the job. We can also offer alternatives like observation-based assessments if needed.

Risks and unknowns

The main risk is that the technology is still in research phase; there are no established standards for biometric-based readiness evaluation. Adoption may stall if construction firms are reluctant to invest in unproven methods. Unknowns include the reliability of biometric measures in real-world conditions and the willingness of workers to wear sensors.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-9528B2CFF302	2026-01-01	openalex.org	1	Exploring Organizational and Individual Determinants of Construction Workers' Safety Behavior: ...
SIG-A33E4A343E60	2025-12-17	openalex.org	1	Immersive technologies for evaluating industrial safety training in high-risk environments: a r...
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-6ADB457BBC36	2025-11-19	openalex.org	1	A Machine Learning Approach to Predicting Air Quality and Health Risks for Construction Worker ...
SIG-3E11F4C71251	2025-11-13	openalex.org	1	Artificial Intelligence (AI) in Construction Safety: A Systematic Literature Review
SIG-97250BAF9D1E	2025-10-07	openalex.org	1	Investigating advanced technologies to enhance worker safety in hazardous mining environments: ...
SIG-8BC657BF5DA5	2025-10-04	openalex.org	1	Assessing workers' neuro-physiological stress responses to augmented reality safety warnings in...
SIG-80CAA03177B8	2025-09-26	openalex.org	1	Occupational health risks, safety essentials, and safety beliefs among construction workers in ...
SIG-72037039EB81	2025-09-12	openalex.org	1	Safety Violations of Construction Workers: An Empirical Study
SIG-7ADD7236CDF2	2025-09-01	openalex.org	1	Preliminary Design Guidelines for Evaluating Immersive Industrial Safety Training

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:07+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.